MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL AND AUTOMOBILE			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech/M.Tech./MCA/M.Sc. #**	**Curriculum**	**2018**
Program Name	**Aeronautcal**	**Course Type**	**PE**
Course Name	**Airship Technology**	**Course Code**	**AAE 4047**
Semester	**6**	**Max Marks**	**50**
Date	**22-11-2024**	**Duration**	**3 Hours**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP4 LO	
1A	Explain in detail the methodology applied in the design phase for airships.	5	4	2	4	1,4	2
1B	An airship with a total mass of 42,000 kg is climbing at a constant velocity of 5 m/s. The drag coefficient during the climb is C_D=0.072, the frontal area is 80 m², and the air density at the operating altitude is 1.1 kg/m³. The engines provide power to overcome both the drag and gravitational forces. Calculate the total power required for the steady climb.	3	3	1,2,4	3	1,2,3	3
1C	Illustrate the influence of fineness ratio on Cd_0 of an airship	2	2	2	2	2,4	2
2A	An airship has a volume of 17,000 m^3, a length (l) of 91.44m, and a diameter of 19.81m. The fineness ratio (FR) is 4.5 , and the velocity (V) is 37.44m/s. The total fin area is 139.35 m^2, and the rudder area is 19.51 m^2. The airship is operating at an altitude of 3,000 meters where the air density 0.9093 kg/m^3 and the dynamic viscosity is 1.460 x 10^{-5} kg/m.s. Calculate the zero-lift drag coefficient (C_{D0}) and the total drag area (S_D) for the airship.	5	3	2,4,5	3	1,2,4,6	4
2B	Distinguish between the design and operation of conventional airships and stratospheric airships.	3	5	1,2,4	5	1,4	3
2C	Explain the requirements of envelope design in terms of aerodynamic and structural point of view.	2	3	1,2	3	1,4	2
3A	Explain in detail, with diagrams, how to determine the takeoff and landing distances of an airship. Also, mention the main	5	3	1,2,4	3	1,4	2

	parameters that influence both the takeoff and landing distances of an airship.						
3B	Airship envelope is 41000m^3 at sea level condition with internal gas temperature of 18^{0}C. Outside ambient temperature is 25^{0}C and the ballonet volume is 3000m^3. The airship is directly exposed to sunlight which increasing the internal temperature by 22% if then how much lift is increased due to this?	**3**	4	1,2	4	1,2,3	3
3C	Differentiate the main differences in design methodology of airship and aerostat.	**2**	2	1,2	2	1,4	2
4A	Determine envelope and ballonet volume for an airship, which has the capability of 4000kg weight and has the ability to fly at an altitude of 1.5 km (σ=density ratio=0.8638). (Assume ballonet volume is 25% of envelope volume and density of buoyant gas is 0.166kg/m^3). a) Determine envelope and ballonet volume at sea level and cruise altitude b) Determine lifting gas volume c) Determine ballonet volume at cruise altitude d) Determine blower weights if the airship needs to come from 1.5km to sea level within 200 seconds. (assume blower operating at 78% capacity)	**5**	3	1,2,4	3	1,2,3	3
4B	Define the purpose of applying coatings or films on an airship structure. Also, elaborate in detail on the types of materials used for coatings and their properties.	**3**	5	1,2,4	5	1,3	2
4C	Describe types of mooring mast using with various airship on the ground.	**2**	1	1	1	1	2
5A	An airship with following parameters, deteremine speed, power required, and maximum (L/D) at minimum drag. Also analysis range and endurance for constant C_L and constant speed strategies. Volume = 1,350,000ft^3, C_{D0} = 0.032, K=0.29, Gross weight = 121,000lb, BR= 0.8, Fuel weight = 12,100lb, W_{H0} = 32,100lb, W_{H1} = 13,100lb, Speed = 55fps, η = 0.7, BSFC = 0.45	**5**	3	1,2,4	3	1,4	3
5B	Determine the followings: a) Calculate the buoyant force for 4.5m^3 of helium (density = 0.167kg/m^3) at sea level conditions. b) Consider an elastic container with helium of volume 90m^3 at internal gas temperature of 19^{0}C. If the internal temperature is increased by 15% then how much volume of gas will expands, also determine the change in density.	**3**	2	1,3	2	1,4	2
5C	Define factor of safety (FOS) in airship technology. Also write down how much is the recommendated value of FOS for airship and stratosphere airship.	**2**	2	1,2	2	4	2

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	2
2	9
3	25
4	8
5	6

BL	Total Marks
2	26
3	24
4	
5	